

A NEW INTERPRETATION OF THE REDSHIFT OBSERVED IN OPTICALLY THIN TRANSITION REGION LINES

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ABSTRACT

We propose that the pervasive redshift observed in transition region spectral lines is caused by downward propagating acoustic waves. The waves are assumed to be generated in the corona as a result of nanoflares or some other form of episodic heating. The dynamic response of a coronal loop to energy released as heat near the loop apex is studied by solving the hydrodynamic equations numerically, consistently including the effects of nonequilibrium ionization on the radiative losses and on the internal energy. We find that the radiative loss curve may change by a factor of 2 during the loop evolution as a result of flows and waves. Spectral line intensities and line profiles are computed in accordance with the numerical model for the resonance lines of C IV, O IV, O VI, and Ne VIII. The three lower temperature lines display an average redshift on the order of 1 km s^{-1} while the high-temperature line (of Ne VIII) displays little average shift. The largest line shifts occur as the loop undergoes transient heating. A simple analytical analysis is performed to isolate the physical effects relevant to the line formation process. We find that the amplitude of the line shift depends on the characteristic time scale for ionization of the radiating ion as well as the global loop parameters; the time scales for loop cooling, the maximum temperature, and the base pressure. The importance of nonequilibrium ionization in the line-forming process will also have impact on the interpretation of the emission measure derived from nonsteady plasmas.

Subject headings: line: formation — radiative transfer — Sun: transition region

1. INTRODUCTION

Observations of spectral lines formed in the solar transition region may provide possibilities of discriminating between mechanisms responsible for heating the solar corona. There are several advantages in using observations of spectral lines formed in this region. Transition region resonance lines lie in the extreme ultraviolet (EUV) where data with spatial resolution of $1''$ (700 km at the Sun), spectral resolution of 0.05 Å and temporal resolution 5 s are available from the High-Resolution Telescope and Spectrograph (Bartoe & Brueckner 1975). As the lines are optically thin, we may forego solving the radiative transport equation in mapping the observed line characteristics to plasma parameters. The transition region has shorter cooling times than the corona and is thus able to rapidly respond to changes in the energy balance. The major difficulties in understanding the observations is the complicated geometry and time variability of the transition region such that a statistical approach to observation analysis must often be followed. The transport of energy into the corona will affect the transition region both as the energy passes upward from the convection zone and as “overflow” energy is transported downward. In this paper we will concentrate on the observational aspects of the transport of energy by downward propagating acoustic waves.

There are two basic coronal heating theories currently in vogue, namely, resonant absorption of magnetohydrodynamic waves and nanoflare heating (see e.g., Ulmschneider, Priest, & Rosner 1991). The studies of resonant absorption have focused on Alfvén waves generated in deeper layers that become trapped in resonant cavities in the corona. The observational signature of this process should show (1) the signature of stand-

ing or propagating Alfvén waves, with periods in the range 5–300 s in order to fit into coronal loops, (2) the signature of any downwardly propagating signal from the dissipation process itself. We expect that the former signature will dominate the line profile in the transition region unless the dissipation mechanism has a time dependence resembling nanoflares as described below.

Nanoflare heating of the corona depends on the low-frequency random motions of the photospheric granulation. The motions of the footpoints of coronal flux tubes by the photospheric plasma continuously deforms the coronal magnetic field. This continual deformation introduces a large number of tangential discontinuities which, upon reaching some critical strength, dissipate through rapid reconnection (Parker 1988). Coronal heating is achieved by the sum of the heat deposited by all these small reconnection events. Each event has a duration on the order of a few seconds and a spatial scale below present observational limits. Parker (1991) estimates the energy of *one* nanoflare to be $\sim 10^{23}$ – 10^{25} ergs. Releasing a large amount of energy in a small volume and in the course of a few seconds will excite several magnetohydrodynamic (MHD) wave modes in the low β coronal plasma. The signature of the nanoflare heating mechanism in transition region lines will consist of the transition region response to the downward propagating modes. This signature will be modulated by the global heating and cooling of the loop as the loop responds to the heat injected by the nanoflares.

The solar chromosphere-corona transition region is highly structured, both spatially and temporally. Intensity observations by Dere, Bartoe, & Brueckner (1984), Porter et al. (1987), and Habbal (1990) indicate that the dominant portion of the transition region radiation arises in regions of variable emission. There are, however, two characteristics of the observations that seem to be invariant from region to region and from

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time to time; the shape of the emission measure curve and the net redshift of spectral lines. It should be mentioned that the redshift decreases toward the limb of the Sun, which indicates that the motions are primarily vertical. The invariant shape of the emission measure curve and the net redshift extend to several other stars (e.g., Jordan & Linsky 1987; Ayres, Jensen, & Engvold 1988).

Analysis of spectral line profiles show that transition region lines display a net redshift (e.g. Doschek, Feldman, & Bohlin 1976; Dere et al. 1984; Athay & Dere 1989; Hassler, Rottman, & Orrall 1991). The observed mean redshift is typically 5 km s⁻¹ for C IV λ 1548 formed at 10⁵ K but varies considerably. The amplitude of the shift is reduced for lines formed at lower temperatures and chromospheric lines display little net shift (see Samain 1991). There are few observations of higher temperature lines but there is evidence that the net redshift is reduced or disappears in the upper transition region. Doschek et al. (1976) report no substantial downflow in O V λ 1218 formed at 2.5 $\times$ 10⁵ K. Since the wavelength determination of this oxygen line may be influenced by Ly α (λ 1216), it is perhaps more significant that Hassler et al. (1991) report a line shift of 0 $\pm$ 4 km s⁻¹ for the Ne VIII λ 770 line formed at 8 $\times$ 10⁵ K. The center-to-limb variation of the redshifts seem to indicate that the motions are primarily vertical, but a horizontal flow component is also present. There is no clear correlation between intensity and line shift and between line width and shift, but there is correlation between the intensity and the line width (Dere et al. 1984).

The existence of the solar wind makes it evident that there is no net flow of mass from the corona toward the chromosphere, and explanations for the net redshifts have invoked selection effects in the radiative or temporal properties of gas flowing up versus gas flowing down. Pneuman & Kopp (1977) and later Athay & Holzer (1982) hypothesized that the redshifts were caused by the return of spicular material. Alternately, the observational signatures have been interpreted as siphon flow through coronal loops (e.g., Boris & Mariska 1982). McClymont & Craig (1986, 1987) and Mariska (1988) showed that asymmetric heating of coronal loops may drive the flow. Both ideas seem compelling; however, the observations reported by Athay & Dere (1989) seem inconsistent with a steady flow of material returning to the chromosphere.

In this paper we propose that the net redshifts may be explained as a consequence of nanoflare coronal heating. Several of the wave modes generated by a nanoflare near the apex of a coronal loop will propagate along the loop and through the transition region. Alfvén waves are noncompressive, and it is thus difficult to separate upward propagating from downward propagating Alfvén waves. This is particularly true if we have observations only in the transition region. We do expect, however, that the presence of Alfvén waves will broaden the transition region spectral lines (see Hansteen & Maltby 1990). On the other hand, propagating compressive waves will cause spectral line asymmetries and line shifts in the "direction" of wave propagation (Erickson & Maltby 1967; McWhirter & Wilson 1974; Byerly, McWhirter, & Wilson 1975; Gouttebroze 1989). Nanoflare-generated waves are of high amplitude and may evolve into shock waves. Shocks lead to changes in line emission strength (e.g., Elzner 1975; Poletto 1979; Flower & Pineau des Forêts 1983) and also alter the line profile (Hansteen 1991). Thus we should expect a net redshift resulting from compressive waves originating in the corona and propagation down through the transition region. The focus of this paper will therefore be on the effects of the com-

pressive waves that result from the episodic release of energy near the apex of coronal loops. For reasons of computational expediency we consider a one-dimensional loop with strong magnetic field and restrict our attention to the slow-mode waves propagating along the magnetic field in this low β plasma. Other wave modes may be important in a further development of this idea.

2. LOOP MODEL

2.1. Equations and Assumptions

The loop into which the nanoflares are introduced consists of a one-dimensional semicircular geometry of length l and constant cross section containing a plasma of hydrogen, helium, carbon, oxygen, and neon at solar (photospheric) abundances (Grevesse 1984). We solve the equations of mass, momentum, and energy conservation to obtain the run of density (ρ), velocity (u), temperature (T), and population densities (n_{ij}) of elements i in ionization states j along the loop as a function of time. The variable z is the length along the loop while t is the time.

We make the assumption that all species have the same flow speed and the same temperature. In a collisionless plasma acoustic waves are heavily Landau damped; however, at the densities ($n_e \geq 10^9$ cm⁻³) and temperatures ($T \leq 2.5 \times 10^6$ K) we find the ion-ion collision frequency ν_{ii} sufficiently larger than the wave frequency to ensure a collisionally dominated plasma. This also applies to the electron-ion collision frequency ν_{ei} ; we may safely assume that ion and electron temperatures are equal on the time scales of interest in this study. A magnetic induction B large enough to ensure a low β plasma in the low-density transition region and corona is assumed, in this case, acoustic waves propagate along the magnetic field and a one-dimensional approach is justified, since we are ignoring the Alfvén and fast modes.

In the observer's frame the ionization rate equation for element i and ionization state j , is written

$$\frac{\partial n_{ij}}{\partial t} + \frac{\partial}{\partial z} (n_{ij} u) = n_e [n_{i,j-1} C_{ij-1} - n_{ij} (C_{ij} + \alpha_{ij}) + n_{i,j+1} \alpha_{i,j+1}] . \quad (1)$$

The ionization and recombination rates are represented, respectively, by C_{ij} and α_{ij} . It is assumed that local processes dominate the ionization state; the rates are mainly functions of temperature except the dielectronic recombination rates which have a slight dependence on electron density, n_e . Photoionizations are ignored, although they presumably play an important role in the ionization state of hydrogen below $T \approx 2 \times 10^4$ K and of helium below $T \approx 8 \times 10^4$ K.

The spontaneous and dielectronic recombination rates are calculated according to Aldrovandi & Pequignot (1973, 1976), the recombination rates of the hydrogen-like ions are computed according to Allen (1973). The collisional ionization rate is obtained from Bell et al. (1982), and the collisional excitation rates are taken from the compendium of Merts et al. (1980).

The equation ensuring the conservation of mass is written

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial z} (\rho u) = 0 . \quad (2)$$

The momentum equation is written

$$\rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial z} = - \frac{\partial}{\partial z} (p + Q) - \rho g_{\parallel} , \quad (3)$$

where $p = nkT_e$ is the gas pressure, and $g_{\parallel}$ the absolute value of the component of gravity parallel with the magnetic field (z). We have also allowed for an “artificial viscosity” term, $\partial Q/\partial z$, which is discussed in the Appendix. The energy equation is written

$$\frac{\partial}{\partial t}(\rho e) + \frac{\partial}{\partial z}(\rho v e) + (p + Q) \frac{\partial u}{\partial z} = -\frac{\partial F_c}{\partial z} + S - L_r. \quad (4)$$

The source and sink terms of the energy equation are determined by the divergence of the conductive flux F_c , the radiative loss per unit volume and time L_r , and the heating rate S . The internal energy e is calculated as the sum of the thermal and internal energy including only ionization states, the contribution from the excitation energy is negligible.

We consider the radiative losses from the transition region to be optically thin. This should be a reasonable assumption with the exception of the radiation loss from neutral hydrogen in the Ly α line. In this paper heat is added to the footpoints of the loop where hydrogen becomes neutral to balance the Ly α radiative losses and maintain a reasonable chromospheric temperature. Kuin & Poland (1991) have presented an alternate approach to treating the Ly α radiative losses without having to solve the transfer equation. We have not used their method, as it does not allow for a consistent treatment of the radiation and the time-dependent ionization state of hydrogen, the importance of which has been pointed out by Carlsson & Stein (1993).

Optically thin radiative losses are computed consistently with the hydrodynamic equations by summing the individual loss rates of the various processes. In the transition region the most important processes are the resonance line radiation of ionized metals, radiative recombinations, and at higher temperature ($T > 10^6$ K) electron-ion free-free radiation. We therefore write the radiative loss rate as

$$L_r = n_e \sum h\nu_{ul} n_l C_{lu} + L_{\text{rec}} + L_{\text{ff}}, \quad (5)$$

where the sum is over all elements and ionization stages, with population densities n_l , transition energies $h\nu_{ul}$, and excitation rates $n_e C_{lu}$. The quantities L_{rec} and L_{ff} represent the losses from recombination and bremsstrahlung. In the present paper the resonance line losses from hydrogen, helium, carbon, oxygen, and neon along with losses from electron-ion bremsstrahlung are included. This gives a fairly good representation of the radiative loss curve for the temperature range 2×10^4 K $< T < 3 \times 10^5$ K. The omission of iron from the radiative losses leads to an underestimate of the losses at temperatures above 3×10^5 K (e.g., Landini & Monsignori-Fossi 1990). This will in turn lead to a greater conductive flux and therefore a larger temperature gradient in the transition region than would be the case were iron included in the radiative losses. We find that the computed radiative loss curve $\Phi(T) \equiv L_r/(n_e n_H)$ varies by more than a factor of 2 in the transition region during the time evolution of one model calculation as a result of nonequilibrium ionization as described in § 5. Figure 1 shows this variation, where the maximum and minimum loss curves $\Phi(T)$ are plotted. This variation is much larger than the 10% gain in accuracy achieved by including ions such as silicon and nitrogen in equilibrium.

The conductive flux F_c is assumed to run parallel with the magnetic field and is computed in this paper according to Spitzer (1962) formula,

$$F_c = -\kappa_0 T^{5/2} dT/dz \quad (6)$$

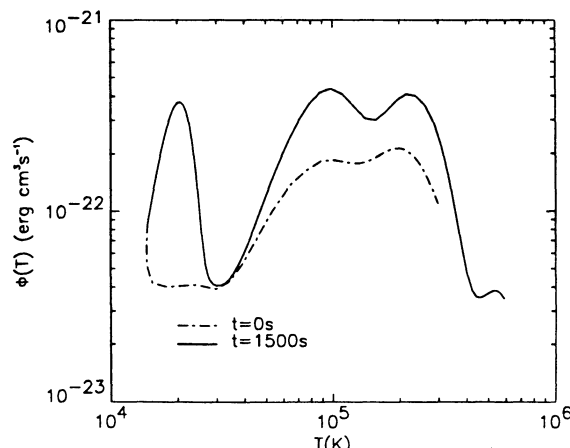


FIG. 1.—Radiative loss curve $\Phi(T)$ as a function of temperature for 2 times in the model calculation. The difference between these times is caused by nonequilibrium ionization as plasma is heated or cools as explained in § 5. Notice the large effect at $T \approx 2 \times 10^4$ K caused by neutral hydrogen flowing up the transition region temperature gradient as chromospheric material is heated, this effect is similar to one seen by Avrett (1991), although there the flow is caused by ambipolar diffusion.

where $\kappa_0 = 1.1 \times 10^{-6}$ ergs $\text{cm}^{-1} \text{s}^{-1} \text{K}^{-7/2}$. This should be an adequate approximation for the coronal and upper transition region part of the loop, but may have to be revised for the lower transition region (e.g., Fontenla, Avrett, & Loeser 1990) or if the temperature gradient becomes extremely large (e.g., Shoub 1982, 1983).

It should be pointed out that the energy balance is coupled to the ionization state of the plasma both through the calculation of the internal energy and through the radiative losses. For several of the relevant ions the characteristic times of collisional ionization and recombination are long compared to the dynamic time scales of the plasma, and we cannot expect the plasma to be in ionizational equilibrium.

2.2. Initial Loop

The initial loop was set up with footpoint pressure of $p_g = 0.7$ dyn cm^{-2} and temperature of $T = 1.5 \times 10^4$ K. Since photoionizations are ignored hydrogen is some 65% neutral at this temperature, giving a footpoint electron density of $n_e = 8 \times 10^{10} \text{ cm}^{-3}$. In the “chromospheric” region stretching up for ≈ 1000 km, heating and radiation are balanced. This is achieved by adding a heating term $S = 3 \times 10^{-23} n_H^2$ to the energy equation; since this scales approximately as the radiative losses (mainly from Ly α) the chromospheric temperature is constant until the transition region temperature rise is encountered. Figure 2 shows the model loop at the onset of the calculation, where the temperature, electron density, and gas pressure are plotted in the left panel, while the velocity is plotted in the right panel. All these variables are plotted as a function of distance z along the loop, which is given a length of $l = 20$ Mm.

The initial transition region and corona lie above the chromosphere with a base pressure of $p_g \approx 0.07$ dyn cm^{-2} and a maximum temperature of $T \approx 3 \times 10^5$. The transition region has a temperature scale height $H_T \approx 100$ km as determined by the divergence of the thermal conduction and the radiative loss terms which dominate the energy equation in this temperature range. The heating function used to heat the chromosphere plays a small role in the transition region and is unimportant in the energy budget at temperatures above $T \approx 3 \times 10^4$ K—

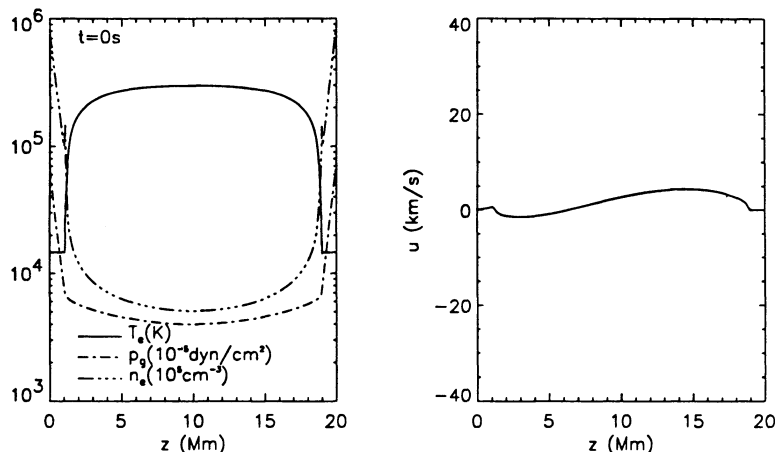


FIG. 2.—The initial distribution of temperature, pressure, and electron density (left panel) and velocity (right panel) with distance (z) along the flux tube. The loop is nearly in force equilibrium, but the apex radiative and conductive losses are not balanced by any heating. The “corona” is condensing onto the chromosphere which is moving up the loop legs. The asymmetry in the velocity is due a low-amplitude acoustic wave which is bouncing between the two transition regions.

indeed other functional forms give similar transition region scale heights as long as the maximal loop temperature is above some $T > 2 \times 10^5$ K. At no point in the time-dependent calculation do we find a transition region with a temperature scale height on the order of the loop length, although we do find that the position of the transition region in the loop can vary on the order of a few chromospheric scale heights as the transition region is subjected to waves, heating, and cooling.

At the temperatures of the initial loop the “corona” is cooling through radiation and through heat conduction to the transition region where the radiative losses are most effective. As a consequence the transition region pressure is depressed below the hydrostatic pressure and gas flows from the chromosphere and from the corona toward the base of the transition region which is moving upward at a rate of $u \approx 1$ km s $^{-1}$. The asymmetry evident in the velocity shown in Figure 2 is due a low-amplitude acoustic wave which is “bouncing” between the transition regions on either end of the loop.

In Figure 3 we show a plot of the initial contribution functions $h\nu n_e A_i n_H x_{ij} C_{lu}$ (see § 3) for the C IV $\lambda 1548$, the O IV $\lambda 790$, and the O VI $\lambda 1037$ spectral lines as a function of length along

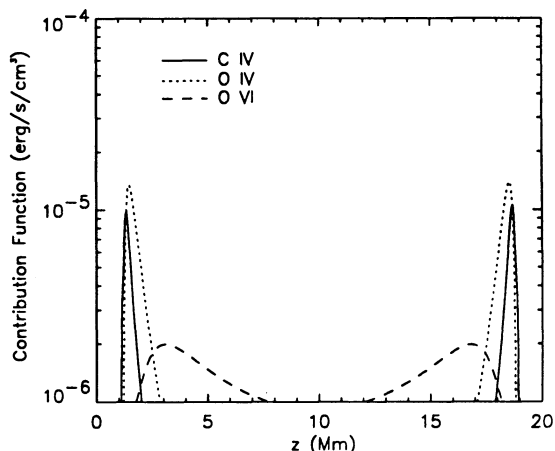


FIG. 3.—Initial contribution function $h\nu n_e A_i n_H x_{ij} C_{lu}$ plotted as a function of length along the coronal flux tube for the resonance lines of C IV, O IV, and O VI. The apex temperature is initially $T \approx 3 \times 10^5$ K while the footpoint temperatures are some $T \approx 1.5 \times 10^4$ K.

the loop. The contribution function for the Ne VIII $\lambda 770$ line is not included as the temperatures are initially too low for significant emission in this line. The C IV and O IV contributions are confined to the transition region and have small extent compared to the loop length. This tendency is enhanced as the temperature gradients become larger in response to loop heating (see, e.g., Figs. 4 and 5). The O VI line initially has a significant contribution from the loop apex but is quickly confined to the transition region as the loop heats. On the other hand, the Ne VIII line has contributions from large portions of the loop during all but the “hottest” periods of the loop evolution. To calculate the intensities, the contribution functions are integrated along the entire loop, and we assume that we are viewing the loop from above when calculating the net line shift; the contributions from both legs of the loop are included and a fill factor $f_x = 1$ is assumed. These assumptions are used throughout this paper when discussing intensities and line shifts. The basic assumptions used in computing the contribution function of the lines are discussed in § 3.

3. LINE FORMATION AND WAVES

We reconsider the emission of optically thin resonance lines from multiple ionized metals formed in the transition region in the presence of atmospheric perturbations. To a good approximation (e.g., Jordan & Wilson 1971) the statistical equilibrium between the upper (u) and lower (l) level of a transition region resonance line is determined locally by the balance between collisional excitation and spontaneous de-excitation

$$n_l n_e C_{lu} = n_u A_{ul}. \quad (7)$$

Here n_l and n_u are the lower level and upper level number densities, respectively, n_e is the electron densities, C_{lu} is the electron impact collision rate, and A_{ul} the Einstein probability for spontaneous deexcitation. Since we are considering resonance lines and $n_e C_{lu} \ll A_{ul}$, we may identify the population of the lower level with the total population of the radiating ion, i.e., $n_l \approx n_{ij}$.

The emergent intensity from the transition region and corona, for an optically thin transition of energy $h\nu_{ul}$ and an emission profile equal to the absorption profile ϕ_v , may be written

$$I_v = \frac{h\nu_{ul}}{4\pi} \int_0^{s_0} \phi_v n_e n_{ij} C_{lu} ds, \quad (8)$$

where we have employed the assumptions above, and we assume an area fill factor $f_x = 1$. The integration is carried out along the line of sight, here represented by path length variable s .

The absorption (and emission) profile is in the case of thermal Doppler broadening given by

$$\phi_\nu(z; \mathbf{n}) = \frac{1}{\pi^{1/2} \Delta\nu_D} \exp \left[- \left(\frac{\Delta\nu - v_u \cos \theta/c}{\Delta\nu_D} \right)^2 \right], \quad (9)$$

where $\Delta\nu = \nu - \nu_{ul}$ is the distance from the rest frequency, and $\theta = \mathbf{e}_z \cdot \mathbf{n}$ is the aspect angle, $\mathbf{n}$ is the direction along the line of sight. All variation of the absorption profile is along the length variable z . The thermal broadening corresponds to a width

$$\Delta\nu_D = \frac{v_{ul}}{c} \sqrt{\frac{2kT_e}{m_A}} \quad (10)$$

where m_A is the mass of the radiating ion.

With a Maxwellian electron distribution function the collision rate is a function of the temperature alone

$$C_{lu}(T_e) = C_0 \frac{1}{g_l} T_e^{-1/2} e^{-h\nu_{ul}/kT_e} \Gamma(T_e), \quad (11)$$

where Γ , the Maxwellian-averaged collision strength (Seaton 1962), is a slowly varying function of the temperature, g_l is the statistical weight of the lower level and $C_0 = 8.63 \times 10^{-6}$.

In order to isolate the physical effects of the compressive wave on the line intensity and line profile, the population of the radiating ion is rewritten as

$$n_{ij} = x_{ij} \frac{n_i}{n_H} n_H, \quad (12)$$

where n_i is the total density of the atom to which the radiating ion belongs, n_i/n_H its abundance relative to hydrogen, n_H the total hydrogen density, and $x_{ij} = n_{ij}/n_i$ is the ionic concentration. The region of line formation is the region along the line of sight where the function

$$g(T) \equiv x_{ij} \exp(-h\nu_{ul}/kT_e) T_e^{-1/2} \quad (13)$$

is significantly different from zero. The function $g(T)$ generally has a sharp maximum near the temperature of maximum ion concentration, and the extent of the region of line formation Δz is thus a good indirect measure of the temperature gradient. This has been used in emission measure analysis of transition region lines (e.g., Pottasch 1964; Jordan & Wilson 1971). Integrating over line frequency and using the expressions given above the total line intensity is seen to vary as

$$I \propto n_e n_H x_{ij} C_{lu} \Delta z, \quad (14)$$

where the variables are evaluated at some appropriate place in the line-forming region.

The response of the optically thin transition region line profile and intensity depends on how the factors described above correlate in the presence of a compressive perturbation. The basic effects may be outlined by considering a sinusoidal progressive sound wave with frequency ω propagating along the loop in the z -direction. The particle velocity u_z varies in phase with the pressure perturbation Δp and the density perturbation $\Delta \rho$

$$\frac{u_z}{c_s} = \frac{\Delta \rho}{\rho} = \frac{\Delta p_m}{\gamma p} \sin \left[\omega \left(\frac{t-z}{c_s} \right) \right], \quad (15)$$

where Δp_m is the amplitude of the pressure perturbation and the sound speed c_s is given by $c_s^2 = \gamma p/\rho$, where γ is the usual ratio of the specific heats (c_p/c_v). The phase relation between the density and velocity perturbation will cause the line intensity to vary with the line-of-sight velocity, leading to a net shift of the spectral line if we integrate over a wave period $P_w = 2\pi/\omega$. Thus, in the absence of other effects we expect redshifted transition region lines if pressure perturbations originate in the corona and propagate toward the chromosphere. On the other hand, *blueshifted* spectral lines are expected should remnants of the acoustic waves which presumably heat the lower chromosphere propagate through the transition region and dominate other pressure perturbations here.

The above discussion was made without reference to departures from ionization equilibrium and variations in the temperature, which will effect both the intensity integrand in equation (8) and the extent of the region of line formation. The effect of nonequilibrium ionization will cause the response of the spectral line to an acoustic perturbation to be not only a function of the density and temperature but also of the velocity and of the history of the plasma. In addition, the waves we are considering are propagating in a time-varying atmosphere as the loop heats and cools. The consequences of these nonlinear effects are taken into account in the numerical calculations.

4. RESULTS

The nanoflare model requires that tangential discontinuities introduced by the random motions of the loop footpoints explosively dissipate, releasing 10^{23} – 10^{25} ergs within a few seconds in a small region. The physics of time-dependent reconnection is a difficult subject, but perturbations introduced by this dissipation are expected to propagate away from the reconnecting region with velocities on the order of the Alfvén speed v_A as well as the sound speed c_s . As noted in § 2.1 the Alfvén and fast mode waves are ignored. In addition to the energy release, momentum will be injected directly to the plasma in the reconnection event. Since we are interested in effects occurring at large distances from the nanoflares we assume that the momentum released is thermalized before reaching scales comparable to that of the numerical grid.

The nanoflares are modeled here by introducing energy in a randomly located computational cell confined to the upper third of the loop, centered on the loop apex. The amount of energy in a single nanoflare is computed from a truncated Gaussian distribution around a chosen average energy, the energy released increasing exponentially within a few seconds from zero to this value. The interval between nanoflares is Poisson-random around a given mean interval between explosions.

Let us now consider the response of transition region spectral lines to the nanoflares released in the upper (coronal) portions of the loop. In the model described below the atmosphere is cycled through a heating phase lasting 120 s where nanoflares are set off in succession, followed by a cooling phase of 250 s where no nanoflares are set off. The individual flare has an average energy release of some $E_{f1} = a e_{f1} \approx 10^7$ ergs cm^{-2} a , where a is the loop cross section. Since we are considering a constant cross section loop we will for simplicity set $a = 1 \text{ cm}^2$ in the following. The nanoflare energy is inserted into the loop as heat. We assume that the heat is deposited in *one* computational cell of length $\Delta z_{f1} = l_g \approx 10^2$ km. Each nanoflare is modeled as having a duration $\tau_{f1} = 5$ s, while the average time between flares is set to 30 s during the heating phase. The

energy inserted is sufficient to maintain a steady state loop of the given length with a maximum temperature of $\sim T \approx 6 \times 10^5$ K.

Releasing a large amount of heat in the corona in the course of a few seconds enhances the temperature locally and the heat conductive flux. Other parts of the flux tube become heated as thermal conduction spreads heat throughout the corona and acoustic waves are excited. The heating causes motions in the transition region as well as perturbations of the transition region temperature gradient and density. This leads to variations in line shifts and line intensities.

The acoustic waves that are generated pass through the transition region and enter the chromospheric footpoints of the loop, eventually leaving the numerical grid through effectively transparent boundaries (see Appendix).

In Figures 4 and 5 we show the line intensity, the line shift and elements of the contribution function for C iv $\lambda 1548$ and O iv $\lambda 779$ as functions of time. In the upper panels the intensity I_ν is plotted as a function of velocity v_{sh} and as a function of time. Overlying this plot the line shift as approximated by the

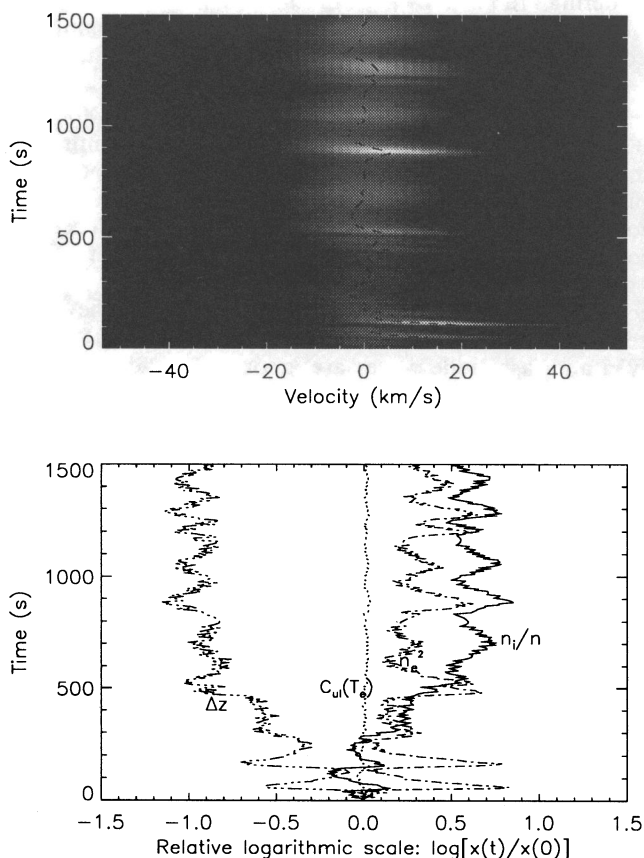


FIG. 4.—Relationship between C iv $\lambda 1548$ intensity, line shift, and elements of the contribution function $\propto n_e^2(n_i/n_i)C_{ii}$, the latter plotted on a logarithmic scale in the lower panel along with the region of line formation (Δz). The total line intensity $\int I_\nu dv$ is approximately given by the product of the contribution function and the region of line formation. In the displayed model, a 120 s period of several nanoflares is followed by a cooling period of 250 s, thereafter the cycle is repeated. The intensity shown is the total from the entire loop as seen from above. Note that C iv formed at 10^5 K is far from ionization equilibrium during most of this period, while O iv (Fig. 5) formed at 1.4×10^5 K is close to equilibrium at all times. Note also that the variation in temperature of line formation ($< 50\%$ for C iv, $< 20\%$ for O iv) matters little for both lines.

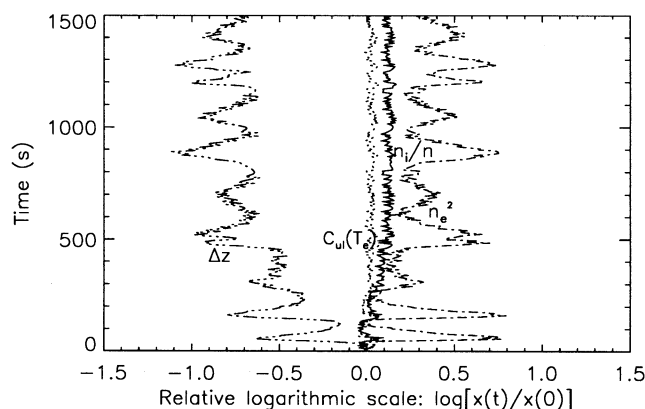
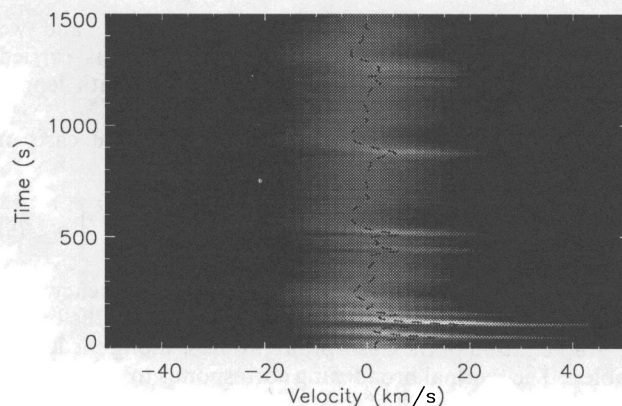


FIG. 5.—Relationship between O iv $\lambda 779$ intensity, line shift, and elements of the contribution function $\propto n_e^2(n_i/n_i)C_{ii}$, the latter plotted on a logarithmic scale in the lower panel along with the region of line formation ($\Delta z \propto 1/\sqrt{T_e}$). Variations in the density are compensated by variations in the region of line formation; as the transition region is pushed into the denser chromosphere by the rising coronal pressure, the enhanced conductive flux from the heated corona forces the temperature gradient to rise, reducing the region of line formation. See Fig. 4 for comparison with C iv. O iv has a fairly complicated term structure and the calculation here should be taken more as indicative of the behavior of a line with short characteristic times for achieving ionizational equilibrium than an accurate portrayal of O iv $\lambda 779$.

first moment,

$$\Delta\lambda_{sh} = \frac{\int I_\lambda(\lambda - \lambda_0)d\lambda}{\int I_\lambda d\lambda}, \quad (16)$$

is plotted after being converted to a velocity scale. Here λ_0 is the rest wavelength of the transition. The relation between the velocity scale and the line shift is $u_{sh} = (\Delta\lambda_{sh}/\lambda)c$. In the lower panels, elements of the contribution function—the electron density squared n_e^2 , the concentration of the radiating ion x_{ij} , the temperature dependent part of the collisional excitation rate $C_{ii} \sim \exp(-h\nu/kT)T_e^{-1/2}$, and the extent of the region of line formation Δz —are plotted relative to their values at $t = 0$ on a logarithmic scale as a function of time. We arbitrarily define Δz as the region where the contribution function is greater than 5% of its maximum value. Both figures show that the model loop may be characterized by three time scales. One time scale of order tens of seconds is associated directly with the compressive waves propagating down from the corona and shows up as “spikes” in the plots, while the other τ_{loop} , which is on the order of several hundred seconds is associated with the global heating and cooling properties of the loop. Also

apparent is the time scale of the heating (120 s) phase when packets of nanoflares occur and the cooling phase (250) when no nanoflares are set off.

Figure 4 shows that the C IV line intensity varies by a large amount as the region of line formation in the transition region responds to the heating and cooling cycle. The line intensity is enhanced and the line is redshifted ($u_{\text{sh}} > 0$) as waves strike the transition region. During the cooling cycle when no nanoflares are set off the wave amplitudes are much smaller and there is a slight blueshift. Thus, there is a tendency for the redshift to be correlated with intensity though the correlation is weak (see, e.g., Fig. 6). The low-frequency behavior of the line intensities is not readily apparent in Figure 4, but will be discussed later. The lower panel of Figure 4 shows that an important contribution to the variation of the C IV $\lambda 1548$ line is caused by variations in the concentration of C IV. Starting from an initial value of $x_{\text{C IV}} \approx 0.1$ the concentration rises almost by an order of magnitude as the loop is heated. The temperature of maximum emission for the $\lambda 1548$ line varies from 9×10^4 K to 1.4×10^5 K. This temperature variation is not large enough to significantly alter the collisional excitation $C_{lu}(T)$. As the nanoflare-induced waves strike the region of line formation the density increases rapidly from the initial value of $n_e = 2.3 \times 10^9 \text{ cm}^{-3}$

and then rapidly relaxes toward its initial value. The electron density also shows variation on the longer time scale of τ_{loop} , by slowly increasing as the loop is heated. This variation in electron density would by itself indicate a large variation in line intensity (see eq. [14]). However, the change in the electron density is partly offset by the variation of the length of the region of line formation, and the product of these two factors varies little. The initial value of the length of the line-forming region for the C IV $\lambda 1548$ line is $\Delta z_0 = 10^3 \text{ km}$.

Figure 5 shows that while the lineshift of the O IV $\lambda 789$ line is similar to that of the C IV $\lambda 1548$ line the intensity variation of the O IV line is smaller than the intensity variation of the C IV line (see Fig. 6). The low-frequency behavior of the O IV line intensity is not readily apparent and will be discussed later. The temperature of maximum emission for this line varies from 1.25×10^5 K to 1.45×10^5 K. This variation is not large enough to significantly affect the collisional excitation rate for O IV. The temporal evolution of the extent of the formation region of the O IV line, which initially is $\Delta z_0 = 1.6 \times 10^3 \text{ km}$, and decreases with time, is partly offset by the evolution of the electron density which initially is given by $n_e = 1.8 \times 10^9 \text{ cm}^{-3}$ and increases with time. The variations in the concentration $x_{\text{O IV}}$ is also small. The product of these factors therefore shows

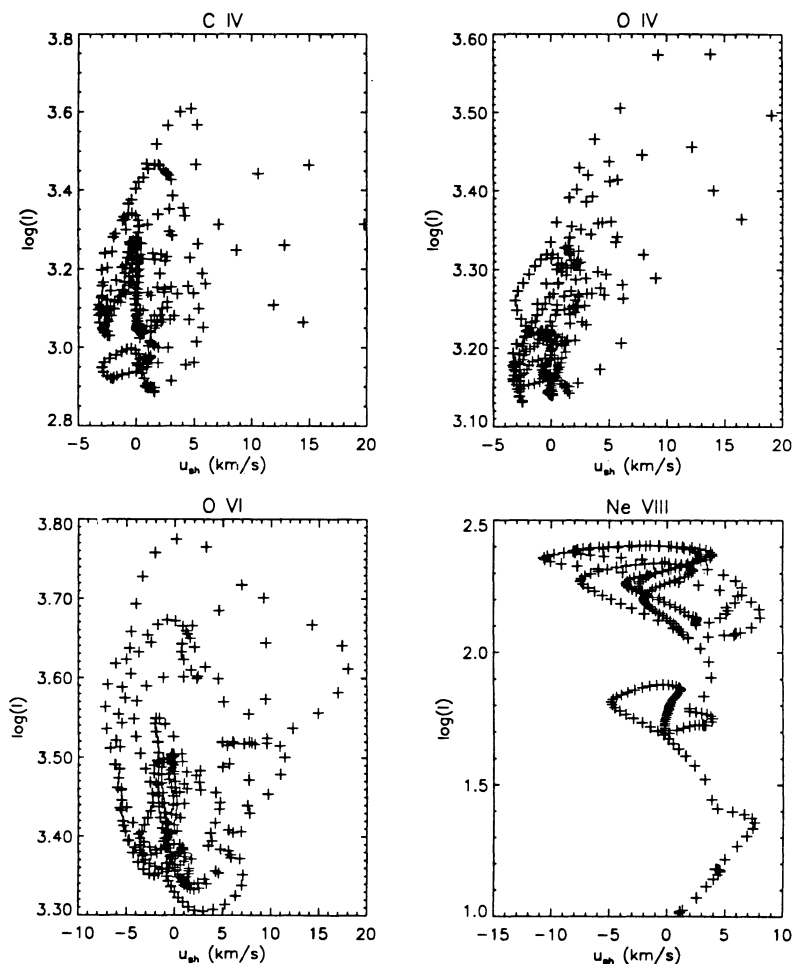


FIG. 6.—Line-of-sight velocity u_{sh} the logarithm of the total intensity (integrated over frequency and loop volume) for the four transition region spectral lines C IV $\lambda 1548$ formed at $T \approx 10^5$ K, O IV $\lambda 789$ formed at $T \approx 1.3 \times 10^5$ K, O VI $\lambda 1037$ formed at $T \approx 2.2 \times 10^5$ K, and Ne VIII $\lambda 770$ formed at $T \approx 8 \times 10^5$ K. One point is plotted for every fifth second in the calculation.

little variation leaving a small intensity variation in the O IV $\lambda 789$ line (see eq. [14]). Despite the small intensity variation in the O IV line, the line does show a net wavelength shift.

In Figure 6 we show scatterplots of the line shift u_{sh} versus the logarithm of the total intensity $\log(I)$, where $I = \int I_\nu(v) dv$, for the resonance lines C IV $\lambda 1548$, O IV $\lambda 789$, O VI $\lambda 1037$, and Ne VIII $\lambda 770$. One point is plotted every fifth second of the simulation. For all the lines except Ne VIII the range in line shifts show a positive asymmetry, which corresponds to a redshift of the lines. The C IV, O IV, and O VI line shifts vary in the range $-5 \text{ km s}^{-1} < u_{sh} < 20 \text{ km s}^{-1}$ while the Ne VIII line shifts are in the range $-10 \text{ km s}^{-1} < u_{sh} < 10 \text{ km s}^{-1}$. The intensity varies by two orders of magnitude for the Ne VIII line as a result of the loop being heated and neon becoming sufficiently ionized to support a significant fraction of Ne VIII. The intensities of the C IV line vary by an order of magnitude, while the intensities of the O IV and O VI lines vary by a factor of 5 or so during the model calculation. The average shifts over the entire calculation, over the first 500 s and the last 1000 s are shown in Table 1. The three lower temperature lines emitted by C IV, formed at $T \approx 10^5 \text{ K}$, by O IV, formed at $T \approx 1.3 \times 10^5 \text{ K}$, and by O VI, formed at $T \approx 2.2 \times 10^5 \text{ K}$, all show net redshifts when they are integrated over the entire model run and during the first 500 s. During the last 1000 s the net shifts

TABLE 1
AVERAGE LINE SHIFTS FOR TIMES ($t_1; t_2$)

Ion	$\bar{u}_{sh}$ (0 s; 1500 s)	$\bar{u}_{sh}$ (0 s; 500 s)	$\bar{u}_{sh}$ (500 s; 1500 s)
C IV	0.9	3.0	0.2
O IV	1.0	2.7	-0.1
O VI	0.5	3.0	-0.6
Ne VIII	-0.8	0.6	-1.0

for the C IV and O IV lines are small and the O VI line shows a small blueshift. The Ne VIII line shows little net shift. The Ne VIII line is formed over an appreciable fraction of the loop length and has intensity contributions from portions of the loop that lie perpendicular to the line of sight. These trends in net line shift are typical also for other model calculations, where we start with hotter loops and used other nanoflare parameters, and other functional forms of the chromospheric heating.

The line widths as approximated by the second moment

$$\Delta\lambda_{wd}^2 = \frac{\int I_\lambda [\lambda - (\lambda_0 + \Delta\lambda_{sh})]^2 d\lambda}{\int I_\lambda d\lambda} \quad (17)$$

are presented in Figure 7, where we plot the line width versus

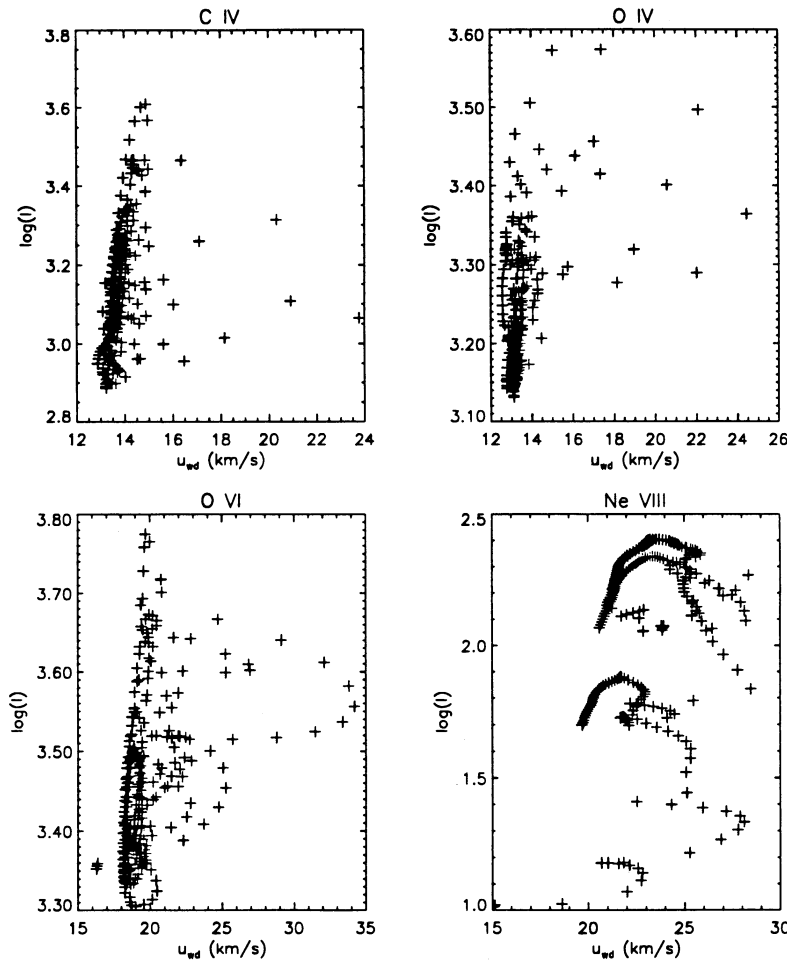


FIG. 7.—Line width u_{wd} vs. the logarithm of the total intensity (integrated over frequency and loop volume) for the following four transition region spectral lines. C IV $\lambda 1548$ formed at $T \approx 10^5 \text{ K}$ with a thermal width at this temperature of $u_{wd,CIV} = 11.8 \text{ km s}^{-1}$; O IV $\lambda 789$ formed at $T \approx 1.3 \times 10^5 \text{ K}$ with thermal width $u_{wd,OIV} = 11.6 \text{ km s}^{-1}$; O VI $\lambda 1037$ formed at $T \approx 2.2 \times 10^5 \text{ K}$ with thermal width $u_{wd,OVI} = 15 \text{ km s}^{-1}$; and Ne VIII $\lambda 770$ formed at $T \approx 8 \times 10^5 \text{ K}$ with thermal width at this temperature of $u_{wd,Ne VIII} = 25 \text{ km s}^{-1}$. One point is plotted for every fifth second in the calculation.

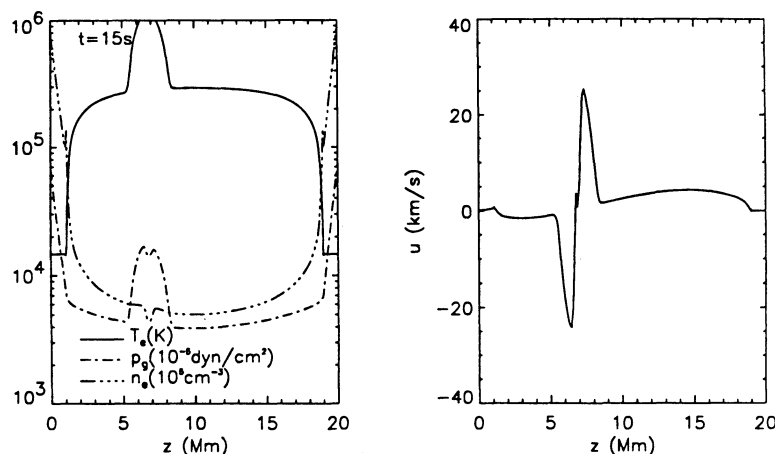


FIG. 8a

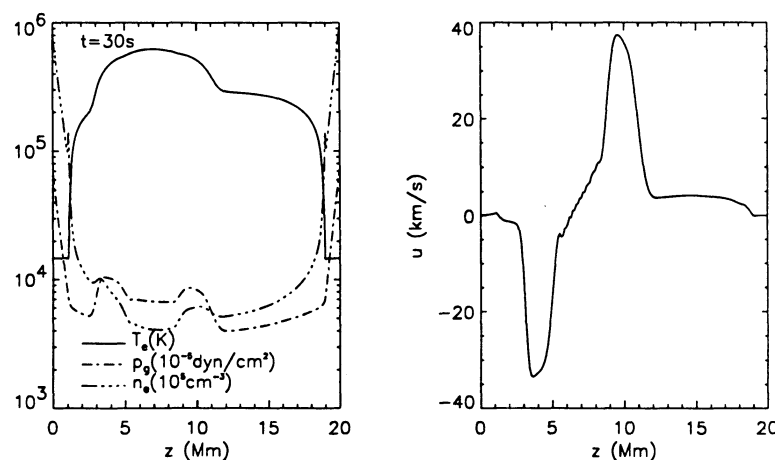


FIG. 8b

FIG. 8.—(a–b) The temperature $T(K)$, electron density $n_e \times 10^5 \text{ (cm}^{-3}\text{)}$, gas pressure $p \times 10^{-5} \text{ (dyn cm}^{-2}\text{)}$ in the left panels and the velocity along the loop $u \text{ (km s}^{-1}\text{)}$ as a function of length along the modeled loop. These snapshots are taken at (a) $t = 15 \text{ s}$ and at (b) $t = 30 \text{ s}$ after the beginning of the run. The first nanoflare started at $t = 11 \text{ s}$. (c–d) The temperature $T(K)$, electron density $n_e \times 10^5 \text{ (cm}^{-3}\text{)}$, gas pressure $p \times 10^{-5} \text{ (dyn cm}^{-2}\text{)}$ in the left panels and the velocity along the loop $u \text{ (km s}^{-1}\text{)}$ as a function of length along the modeled loop. These snapshots are taken at (c) $t = 45 \text{ s}$ and at (d) $t = 60 \text{ s}$ after the beginning of the run. The first nanoflare started at $t = 11 \text{ s}$. Note the wave reflection caused by the transition region temperature gradient in (d). (e) The temperature $T(K)$, electron density $n_e \times 10^5 \text{ (cm}^{-3}\text{)}$, gas pressure $p \times 10^{-5} \text{ (dyn cm}^{-2}\text{)}$ in the left panel and the velocity along the loop $u \text{ (km s}^{-1}\text{)}$ as a function of length along the modeled loop. This snapshot is taken at $t = 75 \text{ s}$ after the beginning of the run, the first nanoflare started at $t = 11 \text{ s}$. Note that the second nanoflare is set off just prior to this snapshot.

the logarithm of the total line intensity for the resonance lines of C IV, O IV, O VI, and Ne VIII. The line widths are converted to a velocity scale $\{u_{\text{wd}} = [(\Delta\lambda_{\text{wd}}^2/\lambda)c]\}$ and are plotted for every fifth second. The line widths are *not* averaged over this time period in order to simulate the effects of finite exposure times. The C IV, O IV, and O VI lines show a slight correlation between width and intensity, although there is some scatter. The Ne VIII widths are at some times smaller than the thermal widths. At these times the maximum loop temperature is below the Ne VIII ionization equilibrium temperature of $T \approx 8 \times 10^5 \text{ K}$ from which the thermal width of the Ne VIII $\lambda 770$ line is computed.

In Figures 8a–8e we illustrate how the loop plasma responds to the energy release by the first and second nanoflares in the model run. The temperature, pressure, electron density, and velocity along the loop as function of position along the loop are plotted. The nanoflares are set off at $t = 11 \text{ s}$ and $t = 65 \text{ s}$. The first nanoflare initially raises the temperature in a small region to above $T = 10^6 \text{ K}$ and generates pressure pulses with $\Delta p/p \approx 1$ which propagate with an amplitude of $30\text{--}40 \text{ km s}^{-1}$

away from the site of the nanoflare in both directions (Fig. 8a–8b). The maximum temperature falls quickly and is on the order of $T \approx 5 \times 10^5 \text{ K}$ some 30 s after the nanoflare. All the plasma behind the pulse is heated to approximately this maximum temperature, forming an expanding bubble of heated gas. At $t \approx 60 \text{ s}$ the pulse strikes the transition region in the footpoint at $z = 0$ (Fig. 8d). The pulse steepens and propagates through the chromosphere, but some of the wave energy is reflected. This is apparent in Figures 8d and 8e which shows a rebound pulse with an amplitude on the order of $\Delta p/p = 0.1$. The second nanoflare, set off at $t \approx 65 \text{ s}$ is visible in the right portion of the loop in Figure 8e. The evolution of the plasma following this second nanoflare is similar to that of the first.

To illustrate the long-term behavior of the loop we plot in Figure 9 the variation of the total C IV $\lambda 1548$ intensity, the average coronal pressure, the column mass of the loop material at temperature above $T = 5 \times 10^4 \text{ K}$, and the loop length measured from the base of the transition region. Initially the transition region radiative losses are too large to be supported by the divergence of the conductive flux. For the first 500 s of

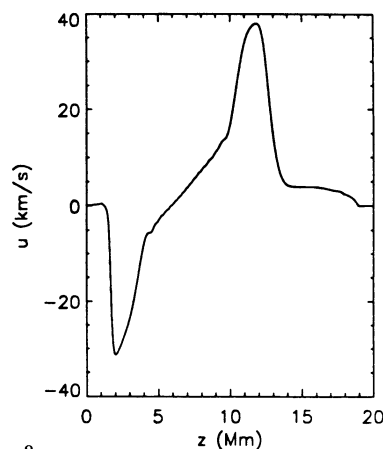
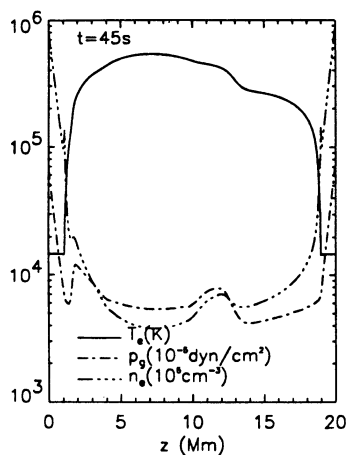


FIG. 8c

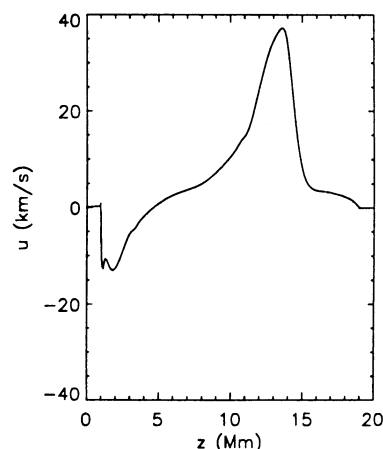
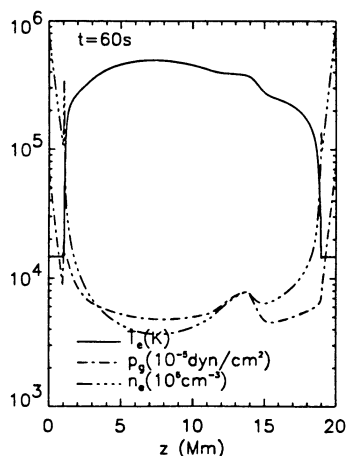


FIG. 8d

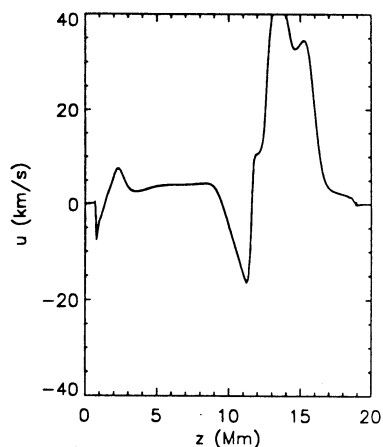
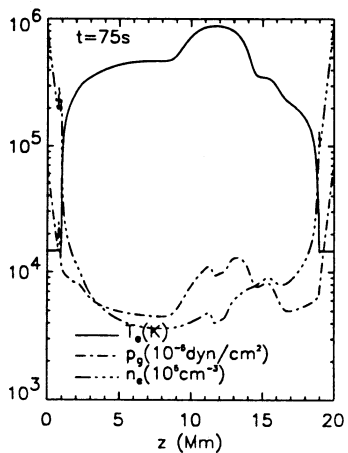


FIG. 8e

the model run material is condensing onto the chromosphere though the loop is heated. As the maximal temperature of the loop becomes high enough around $t \approx 500$ s, the trend is reversed, and material begins evaporating into the corona. This phase lasts the extent of the model run to $t = 1500$ s. The evaporation of chromospheric material leads to an increase of C IV concentration in the loop and thus enhanced C IV $\lambda 1548$ radiation. The O IV $\lambda 789$ line intensity (not shown in Fig. 9) show an increase of only 5% whereas the increase in the C IV

line is 40%. O IV is at all times nearly in ionizational equilibrium, and the density increase in the transition region is offset by the decrease in the extent of the region of line formation which is caused by the increased conductive flux and the enhanced temperature gradient. The loop pressure increases as a result of the coronal temperature increase and, later, as material is brought up into the loop as a result of evaporation. The loop length varies little with a variation on the order of the chromospheric scale height.

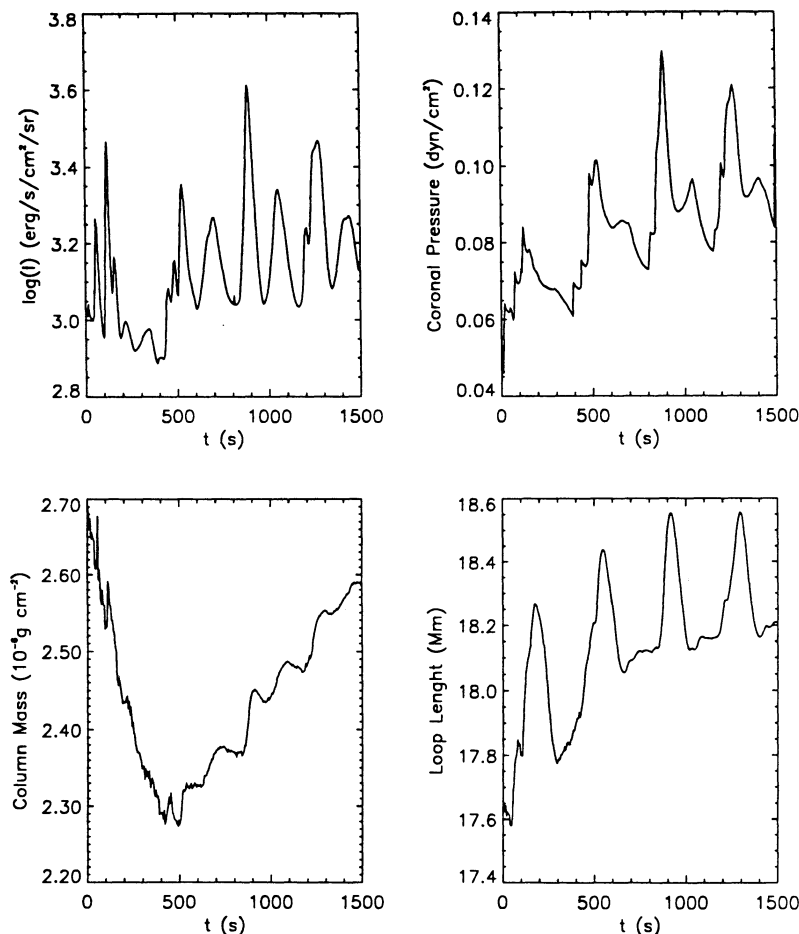


FIG. 9.—The variation of the C IV $\lambda 1548$ total intensity, the average coronal pressure, the column mass (material above $T = 5 \times 10^4$ K), and the loop length with time. During the first 500 s the loop is cooling (condensing) onto the chromosphere. As the loop is heated by nanoflares, the conductive flux originating in the corona becomes large enough to evaporate chromospheric material which flows into the corona. Both condensing and evaporating flows are on the order of $u_{tr} \approx 1 \text{ km s}^{-1}$ in the transition region.

5. DISCUSSION

As nanoflares are inserted into the loop acoustic waves are generated and propagate along the magnetic field. On longer time scales the energy inserted is spread throughout the loop by thermal conduction. Conversely, in a period of no nanoflares the loop cools through thermal conduction and radiative losses. These effects will have an impact on the formation of transition region spectral lines. In this section an attempt is made to isolate the physical processes relevant to line formation through simple analytical arguments.

5.1. Nanoflare-generated Waves

In our calculation a nanoflare of energy $E_{fl} = ae_{fl}$, where a is the loop cross section, generates a thermal wave and an acoustic wave. The thermal wave which initially propagates faster than the acoustic wave spreads the nanoflare energy over a large coronal region. Subsequently as the speed of the thermal wave falls below the speed of sound, the nanoflare energy is transported as an acoustic pulse in both directions away from the site of the nanoflare explosion.

We may estimate the characteristics of the early phases of the thermal wave by assuming that the plasma motion and the density perturbations are unimportant in an energy budget dominated by the conductive flux. As long as thermal conduction dominates the energy balance the energy released in the

nanoflare is conserved within the wavefront defined by the thermal wave. The thermal wave propagates from the position of the nanoflare energy release, z_{fl} , and the temperature behind the wavefront at $z_f = z - z_{fl}$ decreases in order to maintain energy conservation. Under the assumptions written above, the energy released in the nanoflare divided by the (constant) density, and the specific heat at constant volume c_v remains constant:

$$Q \equiv e_{fl}/\rho c_v = \int T dz \approx T z_f. \quad (18)$$

The energy equation for the early phase of the thermal wave is set up following Zel'dovich & Raizer (1967) as

$$\frac{\partial T}{\partial t} = \frac{\partial}{\partial z} \left(\frac{\kappa_0 T^{5/2}}{\rho c_v} \right) \frac{\partial T}{\partial z} \quad (19)$$

by assuming plasma motion and density perturbations are negligible. Replacing $\partial/\partial z$ by $1/z_f$ and $\partial/\partial t$ by $1/t$ allows for an estimate of the velocity of the thermal wave front

$$\begin{aligned} \frac{dz_f}{dt} &\approx \frac{z_f}{t} \approx \frac{\kappa_0}{\rho c_v} Q^{5/2} \frac{1}{z_f^{7/2}} \\ &\approx \frac{\kappa_0}{(\rho c_v)^{7/2}} e_{fl}^{5/2} \frac{1}{z_f^{7/2}}. \end{aligned} \quad (20)$$

The acoustic pulses generated by the nanoflares are of fairly high amplitude, $\Delta p/p \approx 1$, with periods on the order $P_w \approx 10$ s, and wavelengths around $\lambda_w \approx 10^3$ km. These acoustic pulses propagate with a speed on the order of the speed of sound $c_s = (\gamma kT/m)^{1/2}$ in the heated medium. Initially, the thermal wave precedes the acoustic wave, but it is quickly overtaken. With a nanoflare energy of $e_{\text{fl}} = 10^7$ ergs cm^{-2} the thermal wave is overtaken after $\tau_{\text{th}} \approx 10$ s which corresponds to a front position of some $z_f \approx 1000$ km. Thus the thermal wave sets the wavelength and period of the following acoustic pulse, erasing information on the spatial and temporal properties of the nanoflare as long as the energy release occurs in a region smaller than z_f and is of shorter duration than τ_{th} . The acoustic pulse is itself a source of an additional thermal wave as it compresses and heats the coronal plasma. Since the thermal conductivity coefficient is nonlinear, the superposition principle does not hold, and it becomes impossible to separate the thermal and acoustic waves. After a delay of the order of the sound crossing time the transition region is struck by an acoustic shock pulse followed by increased thermal conduction as the flare energy spreads throughout the corona.

The waves propagate toward the transition region where the rapidly decreasing sound speed causes partial reflection of the wave and displacement of the transition region. In the mood presented here we have that typically 70% of the wave energy flux is transmitted while the rest is reflected back into the corona. By considering the amount of momentum deposited in the chromosphere by a pulse, we obtain a transition region displacement of some hundreds of kilometers during a pulse period $P_w = 10$ s. The density and temperature perturbations associated with the pulse passing through the accelerated transition region causes spikes in the contribution function; see equation (14). These spikes show up in the contribution function for the C iv $\lambda 1548$ line in Figure 4 and the O iv $\lambda 789$ line in Figure 5.

5.2. Loop Heating and Cooling

The combined acoustic/thermal wave heats the corona and leads to an increase in the coronal pressure. The transition region does not rebound all the way back to its former position after the wave has passed. The net result of several nanoflares is to drive the transition region into the chromosphere thereby increasing the loop length and the transition region density. At the same time, the temperature gradient in the transition region must increase in order to support the increased thermal conduction from the corona. Let us now consider the low-frequency response of the loop and the line-forming region in the transition region as the loop is heated and cools.

We consider the energy and force balance at a specified temperature in the transition region T_{tr} . The radiative losses may be approximated by $n_e^2 \Phi(T_{\text{tr}}, u)$ where the radiative loss term Φ varies by approximately a factor of 2 as a result of the nonequilibrium ionization described below. The divergence of the conductive flux is roughly given as a function of the loop length l and the maximal coronal temperature T_c , by $\kappa_0 T_c^{7/2}/l^2$. In the upper transition region, the energy balance is dominated by the radiative losses as well as the decrease in thermal conduction flux. The relative size of these terms will determine whether the transition region is being heated or cooled. If the divergence of the thermal conduction flux is larger than the radiative losses at a given temperature material is heated and flows up through the transition region temperature gradient and into the corona. Conversely, an excess of radiative losses at

a given temperature will cause material to cool and flow down. We find that, as the loop heats, energy is deposited closer to the base of the transition region, and the chromospheric plasma is heated. Neutral hydrogen is not immediately ionized as a result of this heating, and radiative losses in the Ly α line from the region where $T \approx 2\text{--}3 \times 10^4$ K are enhanced (see Fig. 1). (The heating included in the model in order to maintain the chromosphere is not insignificant in this region and may therefore play a role in evaporation process.)

The electron density $n_{e,\text{tr}}$ in the transition region will, in tending toward force equilibrium, scale with the coronal pressure; $n_{e,\text{tr}} \approx (p_c/T_{\text{tr}})$. Likewise, changes in the loop length Δl will scale with the coronal pressure such that $\Delta p_c \approx (dp/dz)\Delta l$, where (dp/dz) is the chromospheric pressure gradient. The loop length varies on the order of the chromospheric scale height which is small compared to the loop length. In the context of determining the average divergence of thermal conductive flux we may consider the loop length to be constant. Thus, assuming that the coronal pressure roughly scales with the coronal temperature T_c , we see that for a loop of given length l and base pressure the mass flow is determined by the evolution of the coronal temperature (pressure).

The relevant time scale for the low-frequency transition region motion and flows is given by the characteristic time scale of a coronal loop cooling through thermal conduction

$$\tau_{\text{cool}} \propto \frac{\rho c_v}{\kappa_0 T_c^{5/2}/l^2}, \quad (21)$$

or by the time scale of a loop being heated. The latter is determined by the temporal distribution of "packets" of nanoflares and the time scale of spreading the inserted heat throughout the loop, which is on the order of the sound crossing time. The flows resulting from the heating and cooling cycles of the loop are of fairly small magnitude ($u < 1$ km s^{-1} at $T \approx 10^5$ K) compared to the impulsive velocities generated by the nanoflares. However, the net spectral line shifts associated with these flows weigh heavily in the average net line shift since the line shifts associated with the impulsive nanoflare-generated motion fill a small portion of the total time relative to the flows. In addition, the velocities of the evaporating or condensing plasma are large enough to perturb the ionizational balance and thus the intensity response of transition region spectral lines to variations in the plasma state caused by nanoflare-generated acoustic waves.

5.3. Ionization State

The ionizational state of a radiating ion in a time-varying atmosphere is determined by the characteristic times for recombination and collisional ionization. For ions such as O iv, the rates are sufficiently large that we may consider them in ionizational equilibrium on the time scales of the acoustic waves considered. On the other hand, the helium-like ions C v, O vii, and Ne ix have characteristic ionization and recombination times on the order of 10^2 s. These rates are slow compared to the time scales associated with the dynamics in the transition region and thus lead to ionization states far from equilibrium. This nonequilibrium will, in turn, affect the concentration evolution of the ions "neighboring" the helium-like ions, primarily the lithium-like and, to a lesser extent, the beryllium-like ions. Let us use C iv as an example. Carbon, which is being heated or is flowing up through a temperature gradient from regions of $T < 10^5$ K to regions of $T > 10^5$ K,

will undergo successive ionizations and thus pass from beryllium-like C III through lithium-like C IV and eventually to helium-like C V. The C III is quickly ionized to C IV as the temperature rises but the transition to C V is delayed by slow ionization rates. Thus C IV becomes overabundant at temperatures above its equilibrium temperature. The maximum concentration of C IV is also enhanced above the equilibrium value at $x_{\text{CIV}} \approx 0.4$ to a value of $x_{\text{CIV}} \approx 1.0$. In the reverse process, where plasma is cooling or flowing down to a lower temperature region, the recombination rate of C V to C IV controls the evolution of the ionization state. Since this rate also is slow, C V remains the dominant carbon ion at temperatures far below the equilibrium value. As C V finally recombines, the temperature is low enough that the recombination rate of C IV to C III ensures rapid recombination to C III. Thus, C IV has a maximum abundance at temperatures below the equilibrium temperature of $T \approx 10^5$ K, while the maximum concentration is depressed to $x_{\text{CIV}} \approx 0.1$.

5.4. Summary

In order to achieve an average net line shift in the absence of a net mass flow a correlation between the intensity and velocity is necessary. The model calculations show that we achieve net redshifts in transition region spectral lines as a result of the correlation between the intensity and velocity that occurs in downward propagating acoustic waves. This correlation is enhanced if the loop is cooling and material is condensing onto the chromosphere, while it is suppressed when the loop is heated and chromospheric material evaporates. The ionization state of radiating ions plays an important role in deciding the response of a line to compressive perturbations. This effect discriminates between lines from ions with short and long characteristic times of ionization. The model calculations also show that the computed average net line redshifts are smaller than the observed values. The net redshift in the computed model is for the reasons stated above largest as the loop plasma condenses onto the chromosphere. Increasing the average net shift should therefore be possible in loops where the plasma cools quickly and a large portion of time is spent in the condensing phase. The cooling time of coronal loops decreases with decreasing loop lengths and increasing base densities. In order to match the observed net redshifts, the proposed mechanism of creating average net redshifts may therefore require that most of the transition region emission arises in loops that are shorter than the specific loop considered.

6. CONCLUSIONS

We have considered the hypothesis that compressive waves with origin in coronal nanoflares can account for the pervasive redshift seen in transition region lines. The calculations show that we are able to account for several of the observational characteristics of transition region spectral lines, including the apparently small correlation between intensity and line shift, the range of observed velocities, the correlation between line widths and intensity, and the smaller net shift seen in higher temperature lines. However, the average net shifts calculated in the present model are smaller than the observed values. This problem may be rectified by having the emission predominantly come from loops shorter than the loop considered.

The inclusion of a heating term for the low-temperature "chromospheric" plasma is a tacit assumption that the upper chromosphere and lower transition region are heated by differ-

ent mechanisms than the corona. The role of the chromospheric heating in the model presented here is mainly to set the chromospheric scale height H_p . A secondary role of the chromospheric heating lies in determining the rate of evaporation of chromospheric material as described in § 5. It is thought that the lower chromosphere is heated by the dissipation of acoustic waves. That these waves do not reach the transition region and corona with sufficient amplitude to cause significant heating has been pointed out by Athay & White (1978). In fact, the preponderance of redshifts in the transition region is an indication of the lack of upward propagating acoustic waves. We have made test calculations of acoustic waves propagating *up* from the chromosphere to the transition region. Such waves cause a net blueshift in transition region spectral lines. A complete theory of transition region line shifts should therefore also account for the fate of the acoustic flux presumably heating the lower chromosphere. Here the role played by the ionization of hydrogen and the radiative losses from Ly α are presumably crucial.

The proposed hypothesis to explain the net redshift of transition region spectral lines is an alternative to the hypothesis based on steady flow through loops (e.g., Pneuman & Kopp 1977; Mariska 1988), or to the hypothesis of cooling, falling spicules (e.g., Athay & Holzer 1982; Athay 1984). Separating which of these hypothesis, if any, is correct will require a thorough calculation of the observational consequences of each. A fundamental difficulty lies in determining the geometry of the observed regions. It is important to know if one is observing two or more types of atmosphere at any given time or several "snapshots" of a cycle with period longer than the observational exposure time.

In any case the separation of steady state models from time-dependent episodic models should be fairly straightforward. A gas which is being repeatedly heated and cooled will reflect this through the behavior of emission lines originating from ions with differing characteristic times for achieving ionization equilibrium. Thus similar behavior, from lines with disparate ionization or excitation characteristic times, formed at similar temperatures, such as the C IV $\lambda 1548$ line and the O IV $\lambda 789$ line, will indicate that we are observing steady state loops and favor steady flow models of net redshift. The run of velocity with temperature will also be different for these models and measurements of lines formed at separate temperatures will be able to discriminate between them. Athay & Dere (1989) have made an analysis of transition region line shifts and find that the observed shifts are inconsistent with a continuous flow. Calculations of loop flow by Spadaro, Antiochos, & Mariska (1991) where the effects of nonequilibrium ionization are included also seem to indicate that reproducing the net redshift by considering siphon flow through loops is difficult.

The difference between the spicule and the nanoflare models are somewhat more subtle as both rely on heating and cooling gas cyclicly. The hypotheses differ in where energy is deposited. In spicular scenarios, energy is inserted toward the footpoints of loops, and the net redshift is obtained by considering the time scales of heating and cooling, reasoning that the plasma is longer in the falling, cooling phase than in the rising, heating phase. This asymmetry gives a net observed downflow of the gas at transition region temperature. Emission from C IV favors gas that has a "cold" history. It is not clear that the additional time spent in the cooling phase is enough to cause a net redshift. The calculations presented here show the opposite effect, but it is difficult to discard the spicule hypothesis

without seeing a detailed account of its observational consequences.

Episodically heated loops will, to remain in force and energy balance, adjust their mass content by alternately evaporating chromospheric plasma while being heated and by condensing matter onto the chromosphere while cooling. These processes lead to flows through the transition region. By themselves, these flows will not result in any significant net shift of transition region spectral lines. We have shown that a net redshift is possible to achieve if compressive waves are generated in the corona as a result of the heating by nanoflares. In addition to

generating the slow mode waves studied in this paper, nanoflares will generate Alfvén waves which will further broaden the line profiles. Fast-mode waves may also be generated. These are refracted into regions of low Alfvén speed and may also lead to modifications of the line profiles.

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APPENDIX

1. NUMERICAL METHOD

The equations describing mass, momentum, and energy conservation as well as the ionizational state of the plasma are solved numerically by setting them on a discrete spatial and temporal grid. We have chosen to formulate the equations on an adaptive grid using the Reynolds transport theorem as outlined in Winkler, Norman, & Mihalas (1983). This allows us to resolve the transition region temperature gradient as well as follow any propagating shocks or discontinuities in the solution. According to this formulation equations (1)–(4) may, in general, be written as

$$d_t(f\Delta z) + d_z(fu_{\text{rel}}) + (S\Delta z)^{\text{av}} = 0. \quad (\text{A1})$$

Here f represents the conserved quantity, S any sources or sinks, $u_{\text{rel}} = u - u_{\text{grid}}$ the difference between the material velocity and the grid velocity, while Δz is the size of a computational cell. The operators d_t and d_z work on the discrete quantities g_i^n at time step n and grid position i so that

$$d_t \equiv (g_i^{n+1} - g_i^n)/\Delta t, \quad d_z \equiv (g_{i+1}^{\text{av}} - g_i^{\text{av}}), \quad (\text{A2})$$

where Δt is the length of the time step. The averages appearing in equations (A1) and (A2) are computed according to the Crank-Nicholson (or “ θ ”) method so that at grid position i

$$g_i^{\text{av}} = \theta g_i^{n+1} + (1 - \theta)g_i^n, \quad 0 \leq \theta \leq 1, \quad (\text{A3})$$

when computing the solution for time step $n + 1$. In order to maintain accuracy and stability particle and momentum fluxes (fu_{rel}) are computed by the second-order upwind method of Van Leer (1974). This entails that the equations couple to the two nearest neighboring grid points on either side, leaving a block pentadiagonal system to be solved.

The grid density “ ρ_z ” (number of grid points per unit distance) is computed by the method of Dorfi & Drury (1987), which essentially sets

$$\rho_z(z) \propto R(z) \equiv \sqrt{1 + \left(\frac{z_{\text{av}}}{f_{\text{av}}} \frac{df}{dz} \right)^2}, \quad (\text{A4})$$

where R is the required resolution, f again represents any quantity, z the distance and the subscript “av” some appropriate average of the subscripted variable. In practice, we include information on the variation of the second derivatives and the maximal allowed variation of the grid density with respect to space and time. Solving this equation simultaneously with the dynamic equations allows us to resolve and follow large discontinuities in the solutions as well as to resolve the transition region temperature gradient.

The conservation equations along with the grid equation are solved simultaneously. For $\theta \neq 0$ the method used is implicit, and we solve this coupled set of nonlinear equations at each time step by the Newton-Raphson method, iterating until the calculated correction to our solution is less than some preset value—typically at a relative correction of 10^{-4} .

To ensure a continuous solution through the shocks artificial viscosity is introduced in equation (3). Following von Neumann & Richtmyer (1950) we write

$$Q = \begin{cases} \frac{4}{3} \rho l_g^2 (\partial u / \partial z)^2, & \text{for } (\partial u / \partial z) < 0, \\ 0, & \text{for } (\partial u / \partial z) \geq 0, \end{cases} \quad (\text{A5})$$

where l_g is chosen to be some fraction of the average grid spacing. The artificial viscosity will give a shock thickness of order l_g while reproducing the correct jump in the dynamic variables, as well as the correct propagation speed of the shock. It turned out that in the models presented in this paper the numerical viscosity was unimportant in both the energy and momentum balance.

When the system evolves in time compressive waves are generated. They propagate along the loop and should pass through the boundaries without reflection. We have implanted transparent boundary conditions by the method of Korevaar & Van Leer (1988): the physical variables are extrapolated outside the numerical grid using the characteristic variables. We find that the method gives a reflection of only a few percent of the outgoing wave energy, as long as the thermal conduction flux is not a dominant term in the energy equation. This condition is satisfied in the cool chromospheric plasma at the loop footpoints, and the boundaries do not significantly perturb our solutions.

The rate equations for elements other than hydrogen and helium couple to the dynamic equations mainly through the radiative loss term. A good approximation of the radiative loss term requires that several elements be included, increasing the number of equations to be solved simultaneously from 4, for a time-independent precomputed radiative loss term, to 37, for a plasma consisting of hydrogen, helium, oxygen, carbon, and neon. We therefore solve, each time step, first the dynamic and grid equations along with the rate equations for hydrogen and helium, and thereafter, separately, the rate equations for the remaining elements. The radiative losses based on these ion densities are then computed before the next iteration. To improve the convergence properties of the system, information on the variation of the ion populations with the dynamic variables is included when solving the dynamic equations. We find that 3 to 4 iterations are required at each time step.

The length of each time step is determined on the basis of the last time step, the Courant condition at the boundaries, and the change in the variables from the last time step. We find that the shortest time steps occur when shock waves are passing through the boundaries. In these cases the grid points must move through the pulse structure instead of following it.

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